

**MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY**

**A TWO-TIER COGNITIVE ARCHITECTURE FOR
AUTOMATED THREAT DETECTION AND RESPONSE
USING AGENTIC AI**

by

Nguyen Duc Binh

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

**MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY**

**A TWO-TIER COGNITIVE ARCHITECTURE FOR
AUTOMATED THREAT DETECTION AND RESPONSE
USING AGENTIC AI**

by

Nguyen Duc Binh

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

Supervisor:
Dr. Bui Van Hieu
Dr. Dang Van Hieu

Abstract

In the era of digital transformation and burgeoning network traffic, modern Security Operations Centres (SOCs) face severe alert fatigue, as the volume of telemetry collected from monitoring systems far exceeds manual analysis capacity. Traditional defense tools exhibit clear limitations: static signature rules generate high false-positive rates due to a lack of context interpretation, while pre-programmed SOAR playbooks execute reliably only on anticipated scenarios. Transitioning to Large Language Models (LLMs) enables automated contextual reasoning in natural language; however, operational adoption remains constrained by latency bottlenecks, high computational costs, general adversarial risks (typically log-substrate prompt injections), and AI hallucinations. To address these challenges, this thesis proposes SENTINEL, a cognitive two-tier security architecture operating fully air-gapped, designed on the principle of placing cheap decisions before expensive ones. Tier 1 combines a Welford $\mathcal{O}(1)$ statistical filter and a LightGBM Machine Learning Gate to settle stream noise at wire-speed, offloading 90.6% to 97.5% of telemetry without invoking LLM resources and reducing median end-to-end latency from 17,174.7 ms to 0.88 ms. Tier 2 executes a LangGraph agentic engine integrated with a Dual-RAG hybrid retrieval mechanism (FAISS + BM25 RRF $k = 60$) over 433 MITRE ATT&CK technique entries, while the Awaiting HITL deferral channel reduces analyst workload by 84.24% while retaining a 95.0% recall on genuine threats. Finally, Delimited Nonce Encapsulation provides structural guarantees to robustly withstand 678 adversarial attack samples (including log-substrate prompt injections), and HMAC-SHA256 signature chaining ensures effective tamper-detection for forensic audit trails.

Acknowledgments

The completion of this Master's thesis and the fulfillment of the requirements for the Master of Software Engineering degree would not have been possible without the invaluable support, guidance, and encouragement from my advisors, faculty members, colleagues, and family.

First and foremost, I express my deepest gratitude and utmost respect to my academic supervisors, Dr. Bùi Văn Hiệu and Dr. Đặng Văn Hiếu. Through their profound scholarly expertise, precise strategic guidance, and rigorous research standards, my supervisors patiently mentored my critical thinking and inspired my research methodology throughout this thesis journey. Their sharp professional insights have not only refined this work but also provided an invaluable intellectual foundation for my future research in cybersecurity and artificial intelligence.

I respectfully extend my sincere appreciation to the Board of Management and leadership of the FSB Institute of Management & Technology, as well as the faculty of FPT University, for imparting solid theoretical and practical knowledge in a professional research environment. I also sincerely thank the leadership and colleagues at the C12 Department, National Data Center. Their trust, flexible accommodations, and rich practical domain expertise provided deep operational insights that substantially contributed to the design, experimentation, and validation of the SENTINEL system.

Finally, I dedicate my heartfelt gratitude to my beloved family, especially my caring wife and young child. Their silent sacrifices, immense patience, and unwavering encouragement throughout late-night experimental sessions served as my strongest pillar of support. Despite my utmost efforts, limitations may remain; I welcome constructive feedback and guidance from the thesis evaluation committee and esteemed colleagues.

Contents

Abstract	i
Acknowledgments	ii
List of Figures	iv
List of Tables	iv
List of Abbreviations	vi
1 Introduction	1
1.1 Background and Motivation	1
1.2 Objectives, Research Questions, Scope and Contributions	2
1.3 Structure of the Thesis	3
2 Theoretical Background and Literature Review	4
2.1 Current SOC Operations and the Cognitive Bottleneck	4
2.2 Deterministic Rules, Online Statistics, and Tier 1 Machine Learning	5
2.3 Local Large Language Models and Tier 2 Agentic Intelligence	5
2.4 Adversarial AI Security and Log Audit Cryptography	6
2.5 Overview of Related Research and Research Gaps	7
3 System Design and Technical Implementation	9
3.1 Research Methodology	9
3.2 SENTINEL Architecture Overview and SOC Stack Integration	10
3.3 Tier 1: Wire-speed Filter and Machine Learning Gate	11
3.4 Tier 2: LangGraph Cognitive Agent and Threat Memory	13
3.5 Dual-RAG Hybrid Knowledge Retrieval	14
3.6 Adversarial AI Security and Cryptographic Audit Sealing	15
4 Experiments and Evaluation	17
4.1 Stream Offloading Performance and Filtration Economics	17
4.2 Adversarial AI Guardrails and Cryptographic Audit Integrity	21
4.3 Agentic Cognitive Reasoning and Attack Technique Attribution	23
5 Conclusion	28
References	1

List of Figures

Figure 1.	SENTINEL architecture overview. (Source: Author)	11
Figure 2.	Tier 1 detail diagram. (Source: Author)	12
Figure 3.	Tier 2 agent state-graph. (Source: Author)	14
Figure 4.	Dual-RAG hybrid retrieval and output evidence-anchoring guardrail. (Source: Author)	15
Figure 5.	Input Nonce encapsulation and output HMAC audit chaining. (Source: Author)	16
Figure 6.	Cumulative latency distribution over 500 events. Generated from latency_benchmark.json. (Source: Author)	19
Figure 7.	Static pre-filter block rate by sample provenance beside the false-flag rate on benign logs. Gen- erated from robustness_results.json and adversarial_negative_results.json. (Source: Author) . . .	22
Figure 8.	Attribution by technique against the retrieval ceiling. Generated from the three RQ3 result files. (Source: Author)	24
Figure 9.	Alert triage at Tier 2, contrasting the confirmation channel with the deferral channel. Generated from tier2_decision_results.json. (Source: Author)	26
Figure 10.	Left: action accuracy of Tier 1 alone against the complete two-tier system, with 95% confidence intervals. Right: the four judge axes, with context precision the weak- est. Generated from ablation_action_scores.json and reasoning_eval_results.json. (Source: Author)	27

List of Tables

Table 1.	Comparative analysis between SENTINEL and existing security de- fense solutions	8
Table 2.	Per-tier offload rate on two streams with differing attack base rates .	18

Table 3.	Throughput of control configurations on an identical 150-event stratified subset	20
Table 4.	Guardrail layers, measurement denominators, and results	21
Table 5.	Attribution by technique on the 500-sample payload slice, with the retrieval ceiling for the same queries. Attribution is scored on all 250 labelled samples; retrieval on the 243 of them that Tier 1 escalated, and at parent-technique granularity, so the two denominators differ and are not interchangeable.	24

List of Abbreviations

Abbreviation	Full Description
AI	Artificial Intelligence
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CTI	Cyber Threat Intelligence
DSRM	Design Science Research Methodology
ECDF	Empirical Cumulative Distribution Function
FAISS	Facebook AI Similarity Search
FNR	False Negative Rate
FPR	False Positive Rate
FSM	Finite State Machine
GGUF	GPT-Generated Unified Format
HITL	Human-in-the-Loop
HMAC	Hash-based Message Authentication Code
IDS / IPS	Intrusion Detection / Prevention System
KV	Key-Value (LLM KV-Cache Mechanism)
LLM	Large Language Model
MITRE ATT&CK	MITRE Adversarial Tactics, Techniques, and Common Knowledge
NIST	National Institute of Standards and Technology
RAG	Retrieval-Augmented Generation
RRF	Reciprocal Rank Fusion
SIEM	Security Information and Event Management
SOAR	Security Orchestration, Automation, and Response
SOC	Security Operations Center
TPR	True Positive Rate
WAF	Web Application Firewall

Chapter 1

Introduction

This chapter states the research context and central problem of SENTINEL, a cognitive two-tier architecture automating threat detection and incident response in Security Operations Centres (SOCs), together with the objectives, research questions, research object, scope, and scientific contributions of the thesis.

1.1 Background and Motivation

In the era of digital transformation and burgeoning network traffic, modern Security Operations Centres (SOCs) face severe alert fatigue, as daily telemetry ingestion reaching millions of log events far exceeds manual analysis capacity [1]. To avoid missing intrusions, monitoring sensors are configured toward high sensitivity, causing the vast majority of queued alerts to represent benign activity that does not require direct analyst intervention [2, 3]. Consequently, the operational bottleneck of a modern SOC lies not in data ingestion, but in allocating the scarce attention of human analysts.

Three traditional defense layers—firewalls with static signature intrusion detection (IDS/IPS), SIEM platforms correlating telemetry via rule sets, and pre-programmed SOAR playbooks—exhibit clear limitations: they lack dynamic contextual reasoning and execute reliably only on anticipated scenarios [4]. Unresolved events accumulate in manual queues, forcing analysts to perform time-consuming tasks: extracting raw logs, reconstructing source IP behaviors, cross-referencing asset inventories, and mapping attack techniques. This sequential process creates both a time bottleneck causing backlog accumulation and a context bottleneck as deciding evidence remains scattered across unstructured log records.

Transitioning to Large Language Models (LLMs) enables automated contextual reasoning in natural language; however, operational adoption is constrained by three major challenges: (1) multi-second LLM inference latency conflicts with wire-speed data streams [5]; (2) the AI reasoning layer itself becomes vulnerable to log-substrate prompt injection attacks [6], mandating tamper-sealed audit trails; and (3) internal security telemetry must execute fully local (air-gapped) on finite hardware due to strict data privacy requirements [7]. The 4-to-5 order-of-magnitude computational cost gap between traditional statistical rules ($\ll 1$ ms) and LLMs (seconds) necessitates designing SENTINEL on the principle of *placing cheap decisions before expensive ones*: offloading stream noise at wire-speed and invoking LLM reasoning only for events that truly warrant complex analysis.

1.2 Objectives, Research Questions, Scope and Contributions

To address the central problem, the general objective of this thesis is structured into three distinct yet mutually supportive research tracks, mapped to three research questions (RQs) and primary scientific contributions. The complete architectural design and quantitative experimental evaluations are demonstrated in Chapter 4.

The first track focuses on telemetry tiering: designing a stateless dual-stage filtration pipeline at wire-speed to offload telemetry at the cheap computing tier. RQ1 is formulated as follows: *How can a multi-tiered architecture be designed to autonomously offload security telemetry at wire-speed, thereby addressing the latency bottlenecks and resource consumption inherent in Large Language Models?* The corresponding contribution quantifies the principle of placing cheap decisions before expensive ones and empirically demonstrates the relationship between the offload rate and the attack base rate.

The second track focuses on reasoning security and audit integrity: protecting the language model via structural guardrails and sealing forensic records. RQ2 is formulated as follows: *What methodologies and security mechanisms can effectively withstand adversarial risks to protect the AI's reasoning flow, while ensuring the integrity and tamper-evidence of forensic audit trails?* The corresponding contribution introduces Delimited Nonce Encap-

sulation, providing deterministic structural guarantees independent of probabilistic classifiers [8].

The third track focuses on cognitive reasoning and incident triage: building a stateful agentic engine integrated with domain security knowledge. RQ3 is formulated as follows: *How can stateful reasoning capabilities and domain-specific knowledge retrieval be integrated into an AI Agent to automate intrusion analysis, attribute attack techniques, and generate transparent forensic reports?* The corresponding contribution provides a stateful LangGraph agent operating over a STIX-aligned MITRE ATT&CK knowledge base, coupled with a RAG-anchoring guardrail to mitigate hallucinations.

The scope and methodology are bounded as follows: the research domain focuses on automated telemetry analysis, alert triage, and AI reasoning security in SOCs. The research object encompasses high-speed telemetry tiering algorithms (application signature Rule-based, Welford $\mathcal{O}(1)$ statistics coupled with LightGBM Machine Learning Gate), deterministic adversarial defense (Nonce Encapsulation and HMAC-SHA256), and stateful reasoning agents (LangGraph FSM) with hybrid retrieval (Dual-RAG FAISS + BM25 RRF). Experimentally, SENTINEL is evaluated on two benchmark datasets, CSE-CIC-IDS2018 [9] and CSIC 2010 [10], alongside 823 adversarial attack samples (AdvBench [11], deepset [12], jackhhao [13]) deployed on an air-gapped workstation.

1.3 Structure of the Thesis

Chapter 2 systematizes theoretical foundations and surveys related work to establish the research gap. Chapter 3 presents the research methodology and technical design of SENTINEL. Chapter 4 presents quantitative evaluation results organized strictly according to the three research questions above. Chapter 5 concludes with a summary of contributions, discussion of limitations, and directions for future work.

Chapter 2

Theoretical Background and Literature Review

This chapter presents the foundational theoretical principles of this thesis, covering: current SOC operations and the cognitive bottleneck, deterministic rules, online statistics, and Tier 1 machine learning, local large language models and Tier 2 agentic intelligence, adversarial AI security and log audit cryptography, and an overview of related work to delineate the research gap addressed by SENTINEL.

2.1 Current SOC Operations and the Cognitive Bottleneck

This section analyzes the core architectural causes of the alert overload described in Chapter 1. In a standard SOC model, monitoring relies on aggregating raw telemetry from sensors such as Next-Generation Firewalls (NGFW), Intrusion Detection Systems (IDS/IPS), and Endpoint Detection and Response (EDR) into a centralized SIEM platform [1], at volumes far exceeding manual human analysis capacity [14]. The primary architectural bottleneck stems from the operational nature of these sensors: they render decisions predominantly via static signatures or deterministic rule sets. Because the cost of missing an intrusion is severe, operators are forced to lower detection thresholds. However, lowering thresholds in signature-based systems exponentially increases false positive volume in analysis queues without enhancing contextual understanding [2, 3].

At the higher operational layer, Security Orchestration, Automation, and Response (SOAR) platforms attempt to reduce manual workload by executing pre-programmed response playbooks [4]. Nevertheless, SOAR is strictly constrained by playbook rigidity: a playbook triggers successfully only when alert parameters precisely match anticipated scenarios.

For complex multi-stage attack chains, novel behavioral anomalies, or unstructured evidence scattered across multiple records, SIEM, IDS/IPS, and SOAR layers collectively fail to form contextual judgments. Consequently, all unresolved cases escalate to manual analysis queues, transforming human analysts into the sole cognitive bottleneck of the entire security lifecycle.

2.2 Deterministic Rules, Online Statistics, and Tier 1 Machine Learning

Operating at wire-speed imposes strict algorithmic constraints on Tier 1: processing mechanisms must achieve constant $\mathcal{O}(1)$ time and memory complexity. First, a rule-based defense layer uses static set-matching and service-port lookups to instantly eliminate coarse anomalies without consuming computational resources. Next, baseline tracking applies Welford’s incremental recurrence formula [15] to continuously update the running mean μ_k and sum of squared deviations S_k per observation, computing standard deviation $\sigma_k = \sqrt{S_k/(k-1)}$ and dynamic Z-scores without storing raw historical telemetry:

$$\mu_k = \mu_{k-1} + \frac{x_k - \mu_{k-1}}{k}, \quad S_k = S_{k-1} + (x_k - \mu_{k-1})(x_k - \mu_k) \quad (2.1)$$

Finally, Gradient Boosted Decision Tree (GBDT) machine learning models—exemplified by LightGBM [16]—provide rapid classification of multi-dimensional telemetry patterns. Combining deterministic rule filtering, Z-score statistical normalization, and machine learning classification creates a high-speed multi-stage defense line, ensuring high-throughput offloading while maintaining robustness against adversarial telemetry variants.

2.3 Local Large Language Models and Tier 2 Agentic Intelligence

Executing the reasoning tier entirely on-premise is mandatory to guarantee internal security data sovereignty, while imposing strict GPU memory boundaries. Quantization

compresses weight representations onto a coarser integer grid, trading a minor amount of numerical fidelity for significant memory reduction [17]. Under GGUF Q4_K_M, the security-specialized Foundation-Sec-8B-Instruct model [18, 19] occupies approximately 7–8 GB VRAM, preserving sufficient headroom for context buffers and retrieval indices on a local GPU.

The reasoning tier orchestrates analysis flows via a LangGraph state graph [20]. To mitigate hallucinations and ground decisions in verifiable evidence [21], the system combines hybrid retrieval across dense vector search (FAISS) [22] and sparse keyword matching (BM25) [23]. Retrieved candidates are fused by rank position using Reciprocal Rank Fusion (RRF) [24]:

$$RRF_Score(d) = \sum_{m \in M} \frac{1}{k + r_m(d)} \quad (2.2)$$

enabling transparent cross-referencing of log telemetry against MITRE ATT&CK technique codes [25] and NIST SP 800-61r2 incident handling guidelines [26].

2.4 Adversarial AI Security and Log Audit Cryptography

Integrating Large Language Models into security operations introduces log-substrate prompt injection risks [6, 8]. Because raw log content is adversary-controlled, content-matching filters or probabilistic keyword detectors remain vulnerable to novel adversarial variants [27]. This theoretical foundation necessitates protecting the reasoning tier via deterministic structural isolation (Delimited Nonce Encapsulation), establishing an absolute boundary between control instructions and untrusted telemetry.

Regarding forensic audit integrity, evidence records are sealed in sequence using HMAC-SHA256 signature chaining [28] under a local secret key. Theoretical boundaries must be precisely stated: symmetric HMAC encryption guarantees tamper-evidence and modification detection, but does not provide non-repudiation, which requires asymmetric digital signature infrastructure.

2.5 Overview of Related Research and Research Gaps

AI-driven security operations research has evolved across three main architectural paradigms, as surveyed by Zhang et al. [29].

The first group, traditional enterprise defenses, comprises SIEM platforms such as IBM QRadar and SOAR tools like Palo Alto Cortex XSOAR [1, 4]. These systems are complemented by online log parsers like Drain [30], which compress syntax but cannot analyze behavioral context. Operators bear the burden of this rigidity: in interviews with SOC personnel, AlAhmadi et al. [3] identified poor alert quality as the primary driver of fatigue. The core strength of this baseline is low latency, a property SENTINEL’s Tier 1 inherits and refines.

The second group, retrieval-augmented classification agents such as CyberRAG [31], orchestrates domain-tuned classifiers behind an LLM agent, reporting over 94% per-class accuracy on web attacks. While effective at analyzing selected events, this architecture does not address front-end triage, lacking a wire-speed filtering layer to manage incoming event volume.

The third group, multi-agent and governance-focused frameworks, includes AutoBnB-RAG [32], LanG [33], and Policy-Guided Threat Hunting [34]. AutoBnB-RAG evaluates multi-agent incident response in simulated tabletop environments (*Backdoors & Breaches*). LanG utilizes LangGraph with human-in-the-loop checkpoints and dual-layer defense (regex plus Llama Prompt Guard 2 achieving 98.1% F1). Policy-Guided Threat Hunting employs reconstruction autoencoders and deep reinforcement learning prior to LLM contextual analysis.

A precise research gap emerges: cost-tiering exists in Policy-Guided Threat Hunting, prompt defense exists in LanG, and local execution is demonstrated across systems. However, no single architecture unifies: (1) resolving the vast majority of telemetry at a statistical/ML tier with sub-millisecond latency and offload rates modeled by attack base rate; (2) stateful reasoning on a local GPU under strict retrieval grounding rules; and (3) structural

prompt defense paired with HMAC-SHA256 forensic sealing. Table 1 provides a detailed comparative analysis.

Table 1. Comparative analysis between SENTINEL and existing security defense solutions

Evaluation Criteria	SIEM / SOAR (QRadar, XSOAR)	LLM Agents (CyberRAG, LanG, Policy-Guided)	SENTINEL Architecture (Thesis)
Core Technique	Static correlation rules, rigid Python/JS scripts	Agentic RAG, multi-agent orchestration, DRL pre-classification	Rule-based signatures + Welford $\mathcal{O}(1)$ + LightGBM T1, LangGraph Agent T2
Operational Goal	Centralized log aggregation and playbook response	Automated incident investigation via language reasoning	Resolve volume at cheap tier, reserve costly tier for complex cases
Latency & Cost Tiering	Single cheap tier, no context reasoning	Present in select works (DRL pre-classification); offload rate not modeled by attack rate	Two-tier: ≤ 0.88 ms at T1; 90% – 97% offloading before T2
Data Sovereignty	On-premise local	Local deployment reported; hardware constraints not strictly bounded	100% Air-gapped on local GPU
Prompt Defense	Not applicable	Probabilistic classifiers (LanG: regex + Prompt Guard 2, 98.1% F1)	Structural Nonce guarantee, payload-wording independent
Forensic Audit	Standard text logs (susceptible to tampering)	Audit logs recorded (LanG: <code>mcp_audit.db</code>); no cryptographic sealing	HMAC-SHA256 tamper-evident evidence chain

Chapter 3

System Design and Technical Implementation

This chapter details the research methodology, followed by the system design and experimental implementation of SENTINEL: the two-tier processing pipeline, Tier 1 wire-speed filter and Machine Learning Gate, Tier 2 stateful LangGraph agent and Threat Memory, Dual-RAG hybrid knowledge retrieval, and adversarial AI security controls with cryptographic audit sealing.

3.1 Research Methodology

The thesis adopts the Design Science Research Methodology (DSRM) [35]. The core contribution of this research is a technical artefact, evaluated quantitatively through objective empirical metrics rather than purely conceptual arguments. The system is structured upon three core technical pillars:

1. *Rule-based, Statistical, and Machine Learning Pillar*: Provides wire-speed stream filtration via application signature rules (Rule-based), Welford $\mathcal{O}(1)$ online statistics [15], combined with a LightGBM Machine Learning Gate [16].
2. *Agentic Logic Pillar*: Provides deep contextual reasoning through a LangGraph state-graph [20] orchestrating a locally deployed Foundation-Sec-8B-Instruct model [18] over hybrid FAISS/BM25/RRF retrieval [22–24].
3. *Adversarial Security Pillar*: Establishes a robust defensive perimeter against log-substrate prompt injection risks [6, 8] and seals evidence records via HMAC-SHA256 signature chaining [28].

Operational efficacy of the SENTINEL architecture is quantitatively evaluated across five core measurement dimensions: classification quality, processing latency and throughput, adversarial resistance, explanation coherence (scored independently by a judge model), and forensic audit trail integrity. Comprehensive empirical results, comparative baselines, and system limitation verifications are detailed in Chapter 4.

3.2 SENTINEL Architecture Overview and SOC Stack Integration

SENTINEL operates under a cognitive two-tier architecture centered on computational cost-tiering. Raw telemetry from perimeter sensors streams into a Redis Stream queue and routes to Tier 1. Tier 1 serves as a stateless statistical and machine learning processing stage, resolving the majority of traffic volume at wire-speed. Among Tier 1’s action vocabulary, only ESCALATE proceeds to Tier 2, while BLOCK_IP, ALERT, AWAIT_HITL, and DROP terminate at Tier 1. This grounds the offload rate definition used in Chapter 4: an event is offloaded when resolved without triggering a language model invocation.

At Tier 2, the LangGraph agent [20] performs deep contextual reasoning over the Foundation-Sec-8B-Instruct model [18] (GGUF Q4_K_M quantized), integrating historical state from Threat Memory and evidence from Dual-RAG. Model outputs are bounded by a deterministic guardrail enforced to 3 action states (BLOCK_IP, LOG, AWAIT_HITL), with all records from both tiers sealed via HMAC-SHA256 signature chains [28]. SENTINEL operates as cognitive middleware between SIEM collectors and defense enforcement systems (NGFW/SOAR), absorbing stream noise at Tier 1 and providing cited justifications for escalated events at Tier 2, guaranteeing data sovereignty via air-gapped execution on a local GPU [7]. The complete processing flow and component interactions are illustrated in Figure 1.

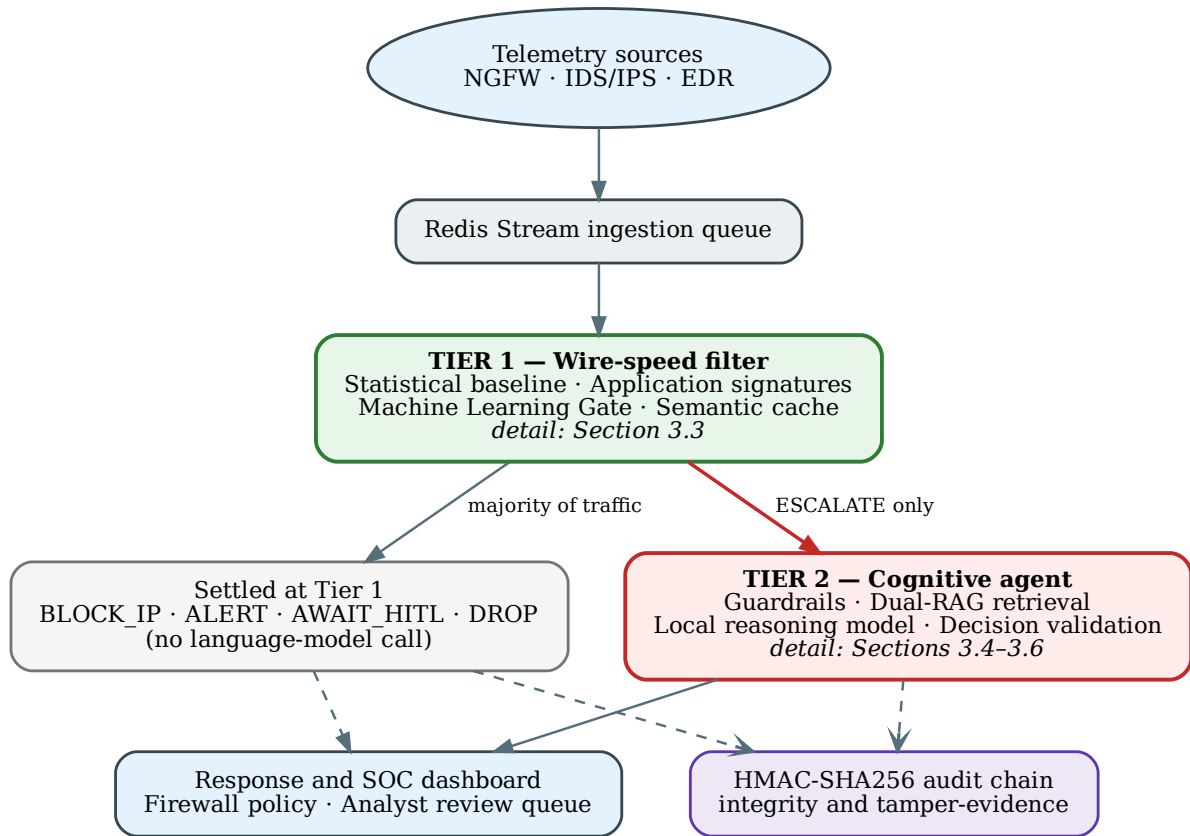


Figure 1. SENTINEL architecture overview. (Source: Author)

3.3 Tier 1: Wire-speed Filter and Machine Learning Gate

Tier 1 operates as a stateless wire-speed perimeter layer combining application-layer signature rules and a Welford $\mathcal{O}(1)$ baseline statistical filter [15] (Equation 2.1). The signature engine covers 30 attack families normalized to OWASP CRS 3.3 benchmarks [36] (23 mapping to OWASP Top 10:2021; 7 targeting network/endpoint anomalies like ransomware and cryptomining). Concurrently, the Welford filter computes streaming Z-scores $Z = (x - \mu)/\sigma$ over flow attributes (packet rate, byte count, duration), updating baseline parameters μ and σ to discard conforming telemetry automatically. Detailed Tier 1 processing flows are shown in Figure 2.

For high-volume repetitive traffic (DDoS, credential stuffing), the Tier 1.75 Semantic

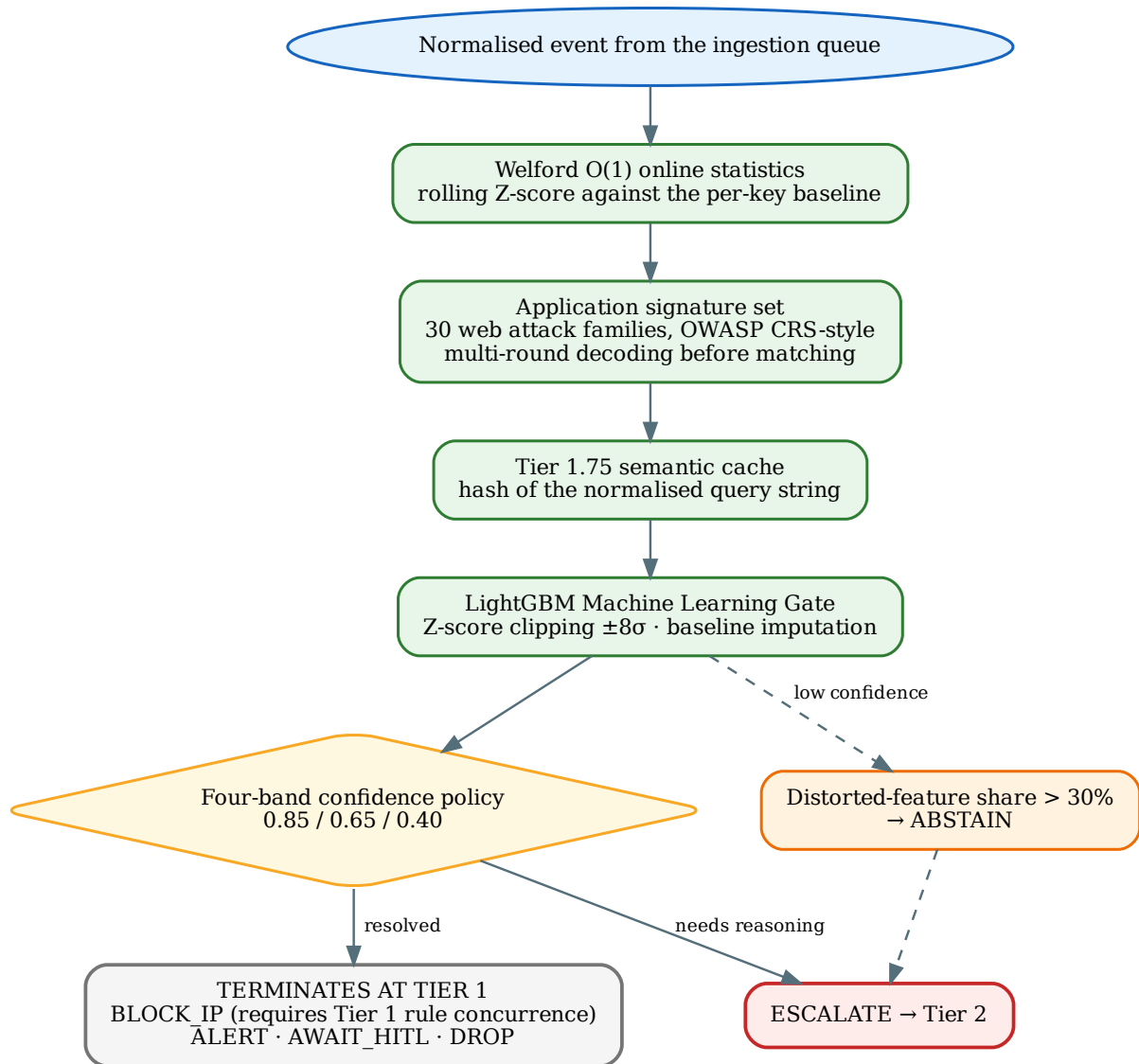


Figure 2. Tier 1 detail diagram. (Source: Author)

Cache hashes normalized query strings in RAM, reusing prior decisions in < 1 ms without recalculation overhead.

Downstream, the LightGBM Machine Learning Gate [16] evaluates non-linear feature interactions, trained on 949,535 NetFlow telemetry records from the CSE-CIC-IDS2018 benchmark [9] (664,674 training, 189,907 test). To counter evasion, the Gate clips Z-scores to $[-8, 8]\sigma$, imputes missing values with baseline means, and abstains to Tier 2 when distorted features exceed 30%. Adjudications map via a 4-band confidence policy (0.85 / 0.65 / 0.40 for BLOCK_IP / ESCALATE / ALERT / LOG), ensuring automated blocking executes only at highest confidence with signature concurrence.

3.4 Tier 2: LangGraph Cognitive Agent and Threat Memory

Tier 2 executes the Foundation-Sec-8B-Instruct model [18] (GGUF Q4_K_M) served by a CUDA-compiled llama.cpp engine on a local GPU. Reasoning flows are orchestrated by a LangGraph state-graph [20] across core stages: guardrail inspection (guardrails), context enrichment (rag_context), incident triage (llm_triage), technique mapping (attack_mapper), and action execution (action_executor) or human-in-the-loop hand-off (human_in_the_loop). The agent state-graph is illustrated in Figure 3.

At attack_mapper, absent explicit knowledge evidence, events route to analyst review (AWAIT_HITL) rather than emitting speculative actions. At llm_triage, ground-truth fields are stripped from the prompt for evaluation objectivity. Input context is managed via Drain log-template mining [30] paired with token budget monitoring.

To correlate multi-stage threat activities, Tier 2 maintains Threat Memory on SQLite (rollback-journal mode preventing Docker file locking). Reputation score S decays over silent period t : $S_{\text{new}} = S_{\text{old}} \times (1 - \lambda)^t$ where λ is the decay rate. Upon reaching $S = 100$, Tier 1 enforces permanent block-on-sight without re-triggering Tier 2. Critical internal infrastructure addresses are whitelisted to eliminate self-inflicted Denial of Service (Self-DoS) risks.

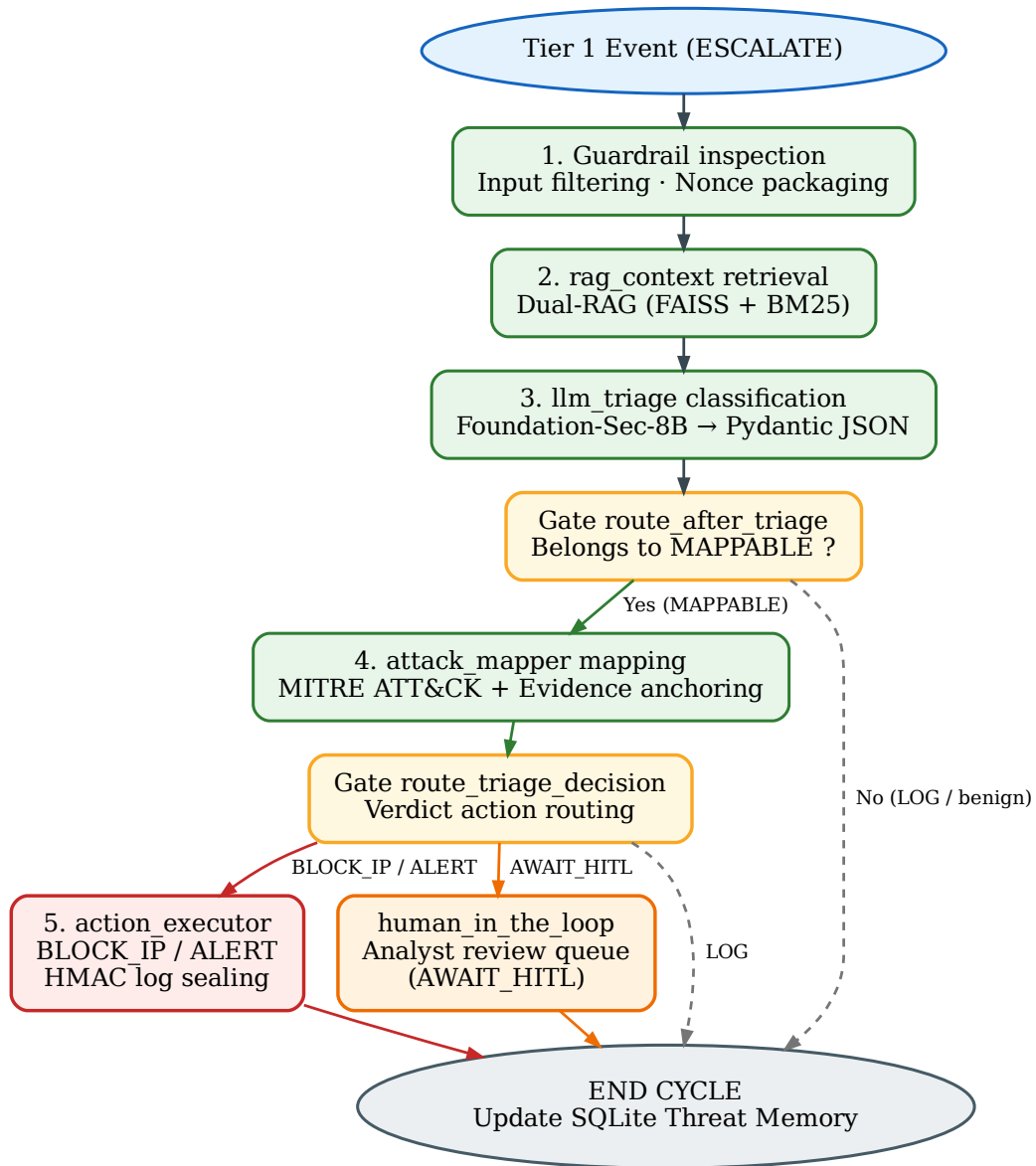


Figure 3. Tier 2 agent state-graph. (Source: Author)

3.5 Dual-RAG Hybrid Knowledge Retrieval

To eliminate hallucinations, Tier 2 retrieves over a knowledge base of 433 STIX 2.1 MITRE ATT&CK entries [25] and NIST SP 800-61r2 guidelines [26]. Parallel channels combine FAISS [22] (Sentence-BERT [37]) for dense vector intent search with BM25 [23] for sparse keyword matching of CVEs and service ports. Results are merged via Reciprocal Rank Fusion (RRF, $k = 60$) [24] and injected at `rag_context` (Figure 4).

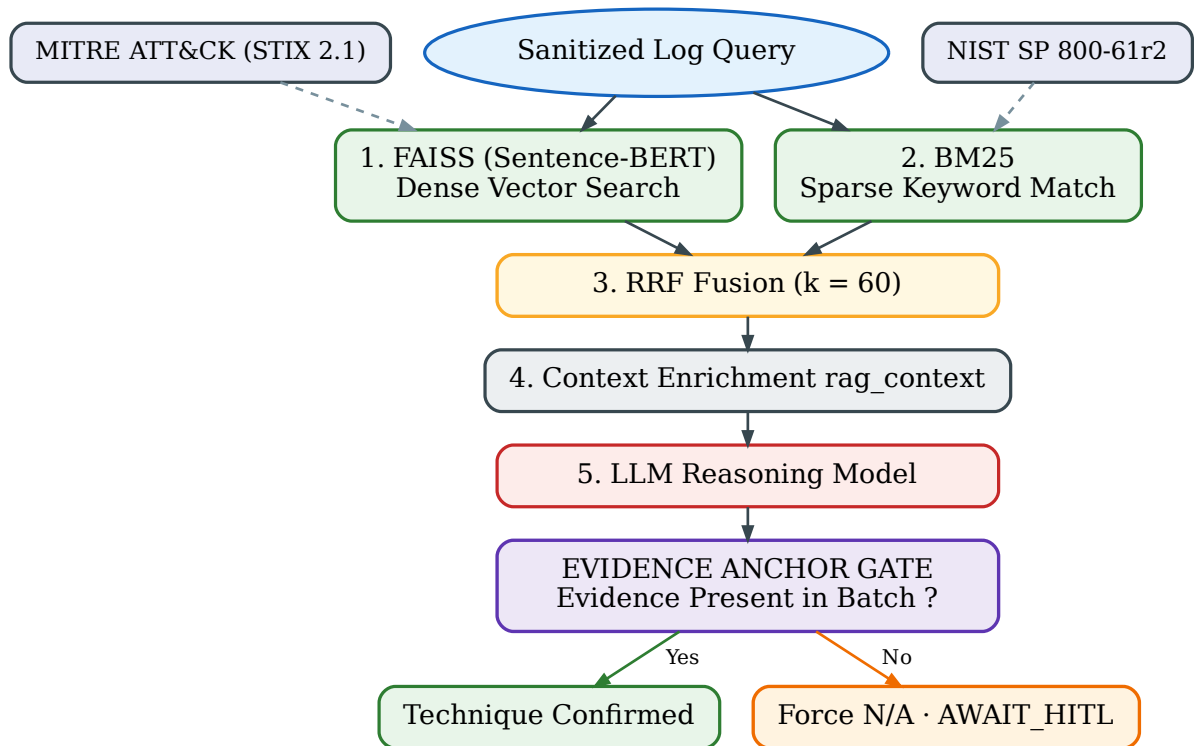


Figure 4. Dual-RAG hybrid retrieval and output evidence-anchoring guardrail. (Source: Author)

Retrieved text is sanitized by Unicode normalization, markup stripping, and command neutralization to prevent indirect injection. At output, an evidence-anchoring guardrail cross-references proposed MITRE ATT&CK codes against batch evidence. Unanchored codes are forced to N/A and deferred to AWAIT_HITL, eliminating hallucination risks.

3.6 Adversarial AI Security and Cryptographic Audit Sealing

Log prompt injection attacks [6, 8] embed control instructions inside raw log fields (User-Agent, URI). SENTINEL mitigates this via Delimited Data Encapsulation using dynamic per-batch random Nonce tokens, preventing delimiter escape (Figure 5). Three deterministic boundary validators guard system integrity: Data (filters malformed logs), Decision (enforces Pydantic LLMDecision and blocks internal IP bans), and Feedback (Zero Trust for AI rules).

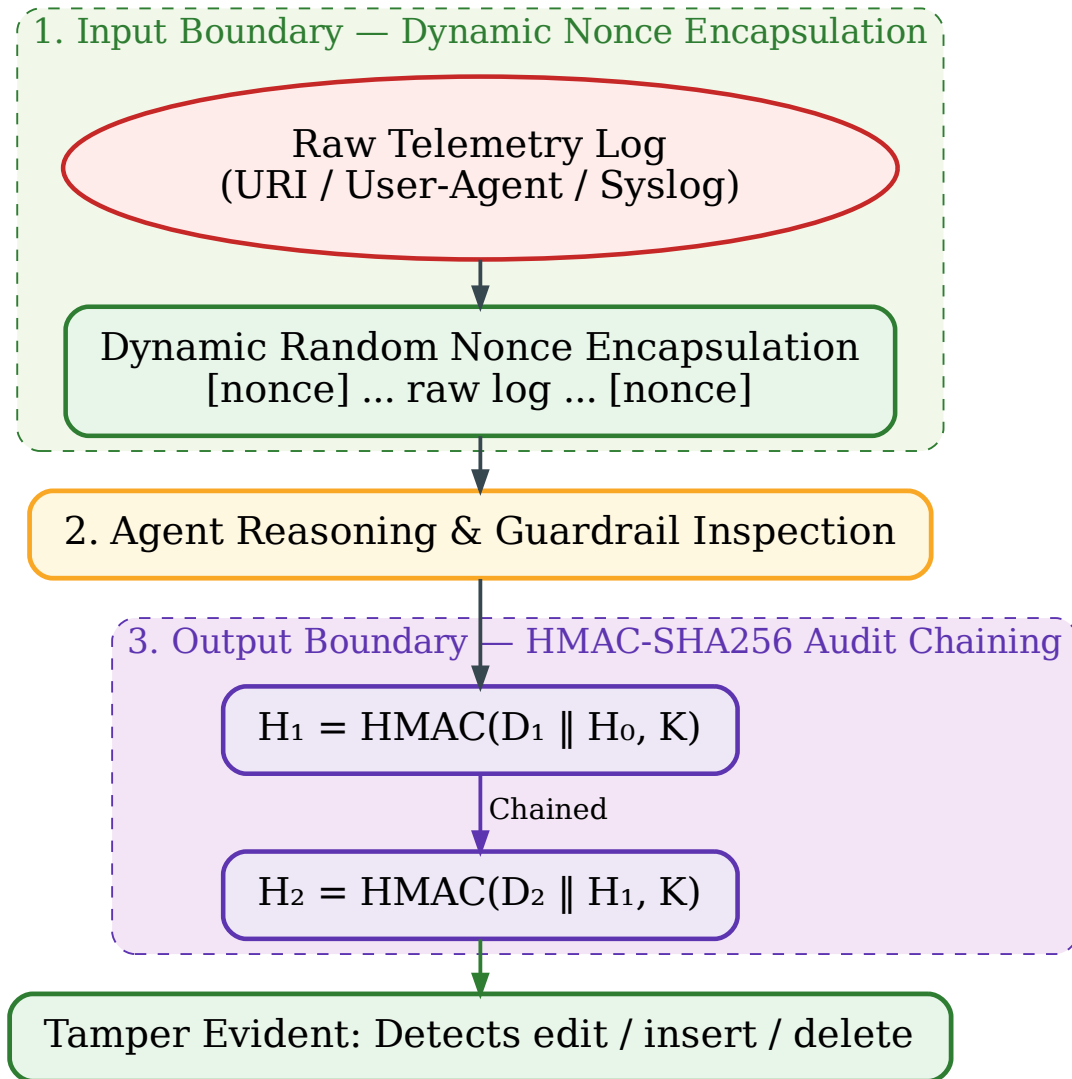


Figure 5. Input Nonce encapsulation and output HMAC audit chaining. (Source: Author)

For forensic auditing, each adjudication record D_i is cryptographically chained using HMAC-SHA256 [28]:

$$H_i = \text{HMAC-SHA256}(D_i \parallel H_{i-1}, K) \quad (3.1)$$

where H_{i-1} is the preceding record hash and K is the local secret key. Modifying, inserting, or deleting records breaks chain continuity and is flagged instantly, ensuring tamper evidence.

Chapter 4

Experiments and Evaluation

This chapter presents the comprehensive empirical evaluation of the SENTINEL system across three core dimensions: wire-speed stream offloading performance, adversarial AI security guardrails with cryptographic audit integrity, and stateful agentic reasoning quality. Experiments were executed on an air-gapped server environment containerized via Docker Compose (Intel Core i7, 32 GB RAM, local NVIDIA GPU running CUDA 11.4.0 with Foundation-Sec-8B-Instruct [18]). Following Section 3.1, offloading performance is evaluated on an operational-scale telemetry stream of 99,717 events with a 9.8% attack base rate reflecting near-realistic SOC conditions [3]. Classification quality and security guardrails are assessed across benchmark datasets CSE-CIC-IDS2018 [9], CSIC 2010 [10], DAPT 2020 [38] and 823 adversarial samples [11–13, 39]. Dynamic rules were frozen during all evaluations, and every ratio is reported with the denominator on which it was measured. Only measurements that substantiate one of the three research questions are reported here; design-parameter verifications and proof-of-concept probes are omitted as they support no contribution claim.

4.1 Stream Offloading Performance and Filtration Economics

Following the definition in Section 3.2, an event is offloaded when it is resolved without invoking the language model, a count that includes BLOCK_IP, ALERT and AWAIT_HITL as well as DROP, none of which consume inference tokens. Table 2 reports the per-tier composition on two streams of differing attack density.

The offload rate behaves as a function of the attack base rate of the stream rather than

Table 2. Per-tier offload rate on two streams with differing attack base rates

Resolving Tier	Benchmark stream, 25,649 ev		SOC-like stream, 99,717 ev	
	Events	Share	Events	Share
Tier 1 (rules + Welford + signatures)	20,663	80.6%	38,207	38.3%
Machine Learning Gate (4-band)	2,568	10.0%	58,992	59.2%
Cumulative — never reaches LLM	23,231	90.6%	97,199	97.5%
Remainder → Tier 2 (LLM)	2,418	9.4%	2,518	2.5%
<i>Attack base rate of the stream</i>		31.6%	9.8%	

as a constant of the system. Within a stream it decomposes linearly as $\text{offload}(p) = (1 - p) \cdot \text{offload}_{\text{benign}} + p \cdot \text{offload}_{\text{attack}}$: benchmark per-label rates of 92.35% and 86.72% reproduce the measured 90.6% exactly, and the corresponding 97.88% and 93.77% reproduce 97.5% on the SOC-like stream. Because the coefficients differ between streams, the relation supports projection within a stream but not transfer between streams, and the 90.6% is therefore not a performance ceiling but the outcome of a benchmark carrying more than three times the realistic attack proportion. The tier bearing the load also inverts, from Tier 1 at 80.6% on the benchmark stream to the Machine Learning Gate at 59.2% on the benign-dominated one, indicating that the two tiers compensate for one another rather than overlap.

A high offload rate is meaningful only if the offloaded portion is adjudicated correctly, since indiscriminate passing would yield an identical headline figure. On `datatest.json` the Gate attains a Matthews Correlation Coefficient (MCC) of 0.667 and a Balanced Accuracy of 0.833; MCC is preferred to Accuracy because a benign-everywhere classifier already scores 90.2% Accuracy on the SOC-like stream. The denominator travels with the metric: of 4,236 events the Gate adjudicates 2,534, as 1,586 lack payload features and 116 are abstained to Tier 2. Its 962 autonomous blocking decisions achieve 100% Precision with zero false positives, an operational prerequisite rather than an incidental result, since blocking legitimate traffic converts efficiency into service disruption. Feature-vector inference completes in 0.287 ms (3,452 events per second), and the Tier 1.75 exact-match cache resolves 81.3% of 1,500 repeated queries at an $8.9\times$ speed-up.

Latency was measured on 500 events sampled at fixed stride from the benchmark stream. Mean end-to-end latency falls from 17,187.9 ms to 5,286.7 ms (-69.2% ; Mann–Whitney U , $p \approx 9 \times 10^{-57}$) and median latency from 17,174.7 ms to 0.88 ms, a factor of approximately 19,500, because the typical event never reaches the model. The 95th percentile is nonetheless worse under the two-tier design, at 25,829 ms against 21,434 ms, since an escalating event incurs the Tier 1 and Gate cost before the model cost: the advantage lies in the body of the distribution rather than the tail, as the Empirical Cumulative Distribution Function (ECDF) in Figure 6 shows. Resource saving is expressed as invocations avoided, 97,199 of 99,717 events, rather than as a VRAM percentage, since no VRAM measurement was performed.

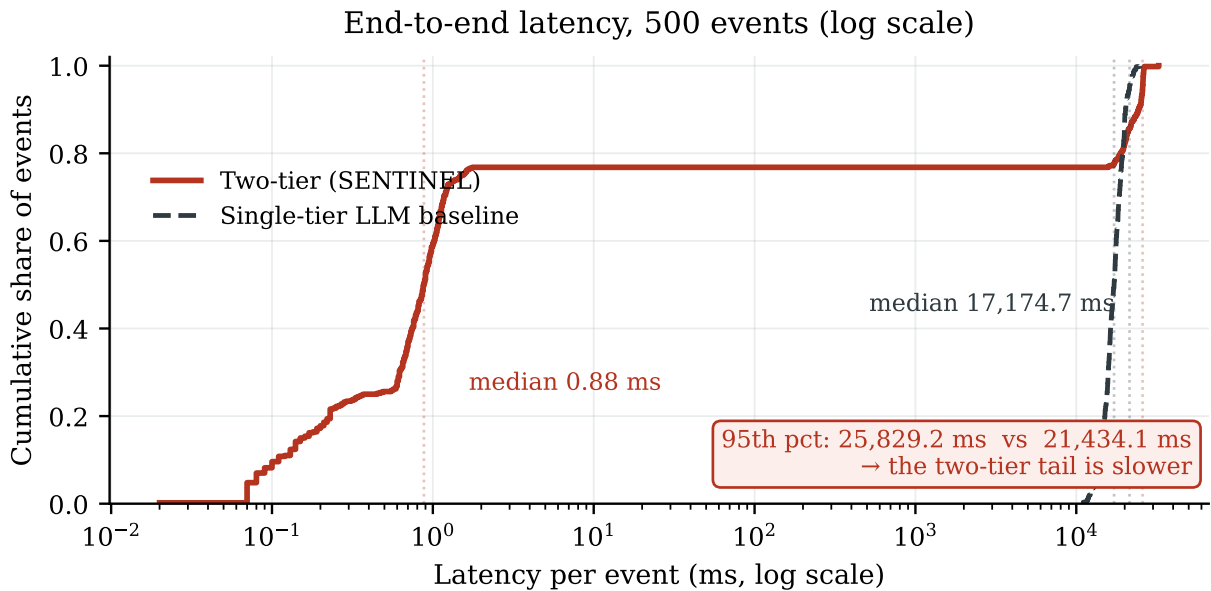


Figure 6. Cumulative latency distribution over 500 events. Generated from `latency_benchmark.json`. (Source: Author)

Four control configurations were measured on one identical 150-event stratified subset rather than by importing published headline figures, which would compare across differing datasets and metric definitions. Table 3 reports the resulting throughput ordering.

The deterministic path executes $20,531\times$ faster than routing every event through the model, while static signatures remain approximately three times faster than SENTINEL, as expected of a layer performing no inference. Two of these figures must not be combined, as the 0.19 ev/s configuration passes every event through the complete Tier 2 agent whereas

Table 3. Throughput of control configurations on an identical 150-event stratified subset

Control Configuration	Throughput
H1 — Static signatures, 30 WAF families	11,129 ev/s
SENTINEL — Tier 1 + ML Gate	3,860 ev/s
H2 — Single-tier LightGBM, flat threshold	2,162 ev/s
H3 — Single-tier LLM (identical data, model, and prompt)	0.19 ev/s

the 17,187.9 ms baseline is a bare model invocation without a token ceiling.

4.2 Adversarial AI Guardrails and Cryptographic Audit Integrity

Research Question 2 (RQ2) treats log-substrate prompt injection as a representative rather than exhaustive risk vector, so two structurally distinct surfaces are evaluated: attack through language, where resistance means an embedded instruction fails to alter the action of the agent, and attack through feature space, where resistance means a distorted sample is not reclassified as benign. Three conventions govern the ratios. The static verdict is the logical disjunction of three detectors recorded individually, so a failure can be localised; a negative control over benign logs is mandatory, since a layer flagging every input also blocks every input; and published-corpus samples are never merged with author-composed ones. Table 4 states each guardrail layer with its measurement denominator.

Table 4. Guardrail layers, measurement denominators, and results

Guardrail layer	Denominator	Result
<i>Tier 1 — attack through language</i>		
Static pre-filter, published corpora	703 samples	132 → 18.8%
by detector: pattern / encoding / delimiter	703	114 / 29 / 0
Static pre-filter, author-composed	80 samples	58 → 72.5%
Static pre-filter, cross-check set	40 samples	2 → 5.0%
Static pre-filter — false flags on benign logs	230 benign logs	46 → 20.0%
<i>Tier 1 — attack through feature space</i>		
ML Gate — evasion, hard mode (all features distorted; 13 flipped, 95% CI 97.86–99.27)	1,036 detected attacks	1,023 → 98.75%
ML Gate — single-feature modes (∞ / extreme)	$1,036 \times 2$	100% both
ML Gate — abstention on distorted input	4,236 events	116 → escalate
<i>Tier 2 — structural guarantee</i>		
Delimited nonce encapsulation	678 hard samples, coverage 100%	678 → 100%
Field-position set, by position	100 published samples	24/24, 23/23, 30/30, 23/23
Schema validation under forced backend failure	1 induced failure	AWAIT_HITL, no crash
<i>Forensic integrity</i>		
HMAC — three core tampering modes	30 trials each	100%
HMAC — negative control (intact chain)	450 records	no false alarm
HMAC — tail truncation	—	0% — undetectable

The static pre-filter blocks 18.8% of the published corpora, which must be read together with the 20.0% false-flag rate measured on 230 benign logs. All 46 false flags originate in the encoding-detection layer, and their operational cost is bounded: a static flag raises the isolation level and annotates the record but neither drops the event nor blocks the source,

making it a rate of unnecessary annotation rather than of legitimate traffic denied service. The author-composed subset blocks at 72.5% because 45 of those 80 samples are encoding-bypass cases the neutraliser was designed to address; the merged rate of 23.3% would overstate efficacy by counting samples authored with prior knowledge of the static rules, so 18.8% is presented as the authoritative metric.

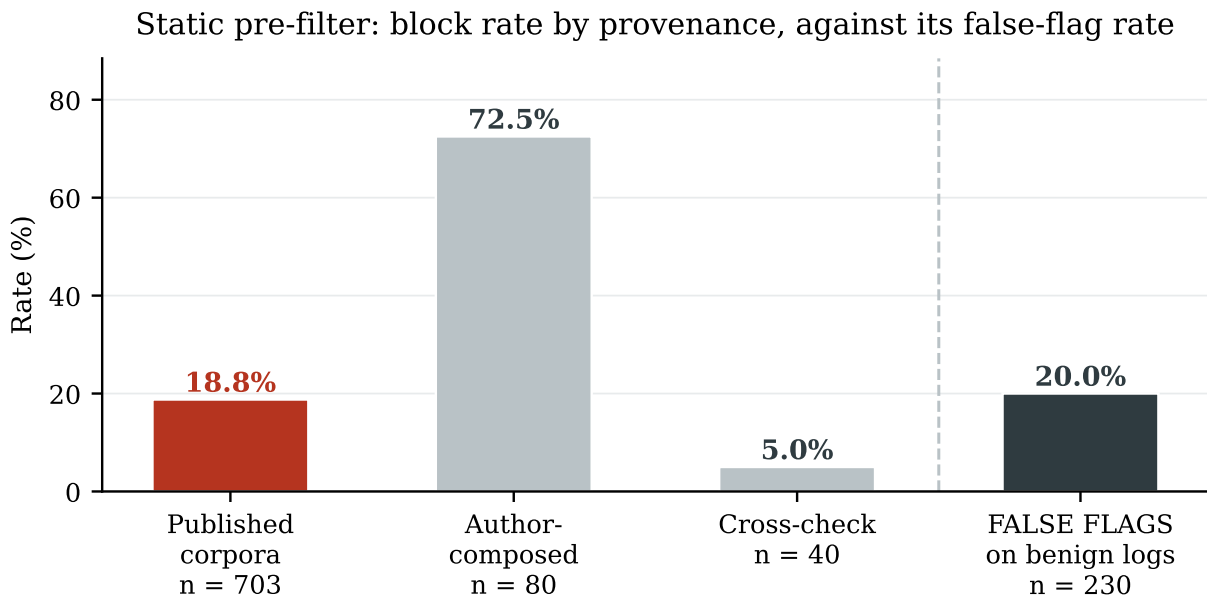


Figure 7. Static pre-filter block rate by sample provenance beside the false-flag rate on benign logs. Generated from `robustness_results.json` and `adversarial_negative_results.json`. (Source: Author)

Under the hard evasion mode, in which every feature of a detected attack is driven to an extreme, 98.75% of samples resist reclassification; the two single-feature modes reach 100%, and the hard mode is cited rather than an average because averaging across two trivially saturated modes would inflate the figure beyond what the defence demonstrates. The structural guarantee derives from Delimited Data Encapsulation, evaluated at 100% coverage over all 678 samples of the seven categories the static layer does not absorb, a superset of the 543 that genuinely bypassed it, so the test is stricter than required rather than favourably sampled. Tier 2 resisted 678 of 678. Unlike a probabilistic classifier, which necessarily leaks whatever it fails to recognise, this guarantee does not depend on recognising malicious content at all. Because encapsulation is applied per field, field position is verified separately: across 100 published samples distributed over URI, User-Agent, message and payload, Tier 2 resisted every sample uniformly across the four positions.

Forensic integrity was verified by recomputing the chain from the genesis record: because each link consumes the hash of its predecessor, a single altered byte in D_j invalidates every link from j onward, rendering detection positional rather than a generic notification that something changed. The three core tampering modes, namely content edit, forged insertion, and deletion of an intermediate record, are each detected in 30 of 30 trials, with no false alarm across 450 intact records. The secret key was loaded from the environment rather than the in-source default, as a key visible in source would permit forgery of the entire chain and invalidate every rate above. Tail truncation remains undetectable, an inherent property of sequential log chaining; the remedy is periodic anchoring of hash values outside the system.

4.3 Agentic Cognitive Reasoning and Attack Technique Attribution

The dimensions evaluated under Research Question 3 (RQ3), namely technique attribution, retrieval efficacy, transparent reporting, and analyst workload reduction, are measured independently because they fail for distinct reasons. Attribution was evaluated on 500 payload-layer samples drawn from CSIC 2010, comprising 250 labelled and 250 benign records. Retrieval alone attributes 80.0% of techniques exactly, whereas the complete agentic pipeline attains 68.0%, with tactic-level accuracy falling from 80.0% to 70.4% and median latency rising from 0.54 ms to 13,293 ms. The reasoning tier therefore costs four orders of magnitude in latency and returns lower attribution accuracy than the retrieval layer it was introduced to improve. Of the 30 samples lost, as many as 22 are cases in which the anchoring guardrail forced N/A because the proposed technique was absent from the retrieved documents of that batch, a deliberate exchange of accuracy for the guarantee that no unsupported identifier is acted upon. Every figure here reads against a majority-class floor rather than zero: T1595.003 accounts for 130 of the 250 labelled samples, so constant prediction already scores 52%.

Table 5 localises the failing component in a way no aggregate can. For T1083 the retriever never returns the correct document, placing the fault in the knowledge base where no down-

Table 5. Attribution by technique on the 500-sample payload slice, with the retrieval ceiling for the same queries. Attribution is scored on all 250 labelled samples; retrieval on the 243 of them that Tier 1 escalated, and at parent-technique granularity, so the two denominators differ and are not interchangeable.

ATT&CK technique	<i>n</i>	Retrieval hit@3	Retrieval only	Full agent
T1190 — Exploit public-facing application	52	1.00	100.0%	63.5%
T1059.007 — JavaScript execution	30	1.00	100.0%	73.3%
T1595.003 — Wordlist scanning	130	0.96	90.8%	86.9%
T1083 — File and directory discovery	12	0.00	0.0%	16.7%
T1071.001 — Web protocols	26	1.00	0.0%	0.0%
All techniques	250	0.93	80.0%	68.0%

stream stage can recover it; for T1071.001 the retriever returns it every time yet attribution remains zero, placing the fault in the mapper. The two require different remedies, and a single aggregate of 68.0% would prescribe neither.

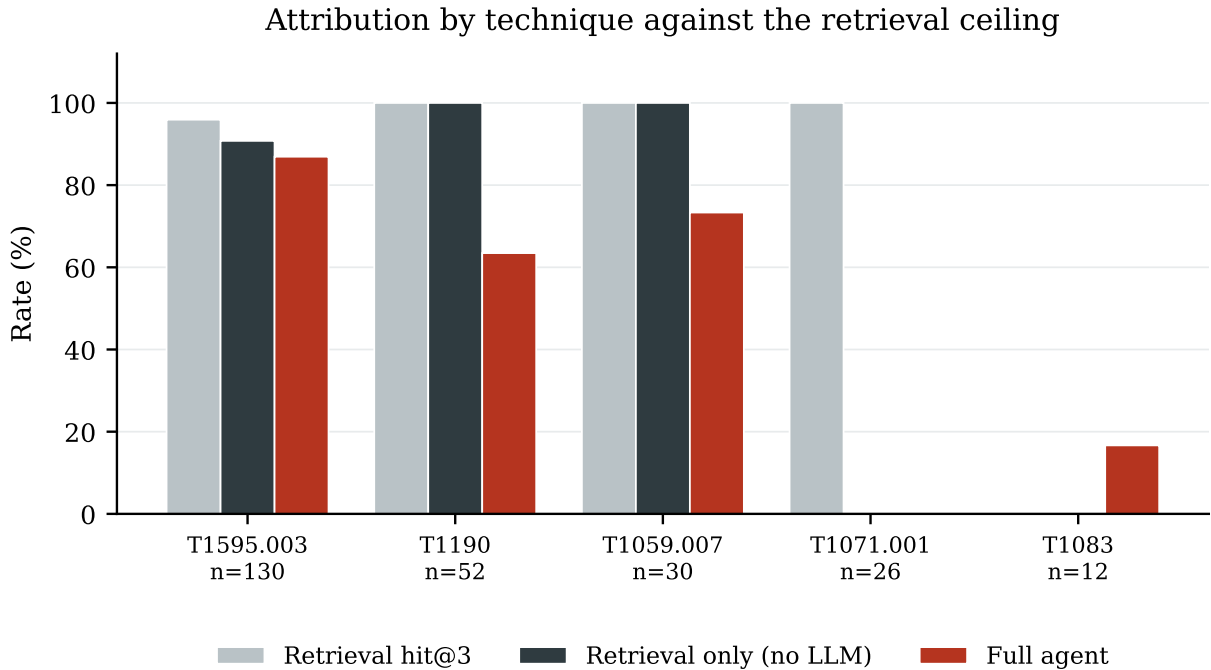


Figure 8. Attribution by technique against the retrieval ceiling. Generated from the three RQ3 result files. (Source: Author)

On the same slice of 243 escalated queries, Dual-RAG attains a Recall@3 of 0.930 (95% CI 0.891–0.956) and a Mean Reciprocal Rank of 0.803, above the attribution figures, confirming that retrieval is not what limits the 80.0%. The unfavourable half is equally explicit: across the full ground_truth set of 1,305 queries Recall@3 falls to 0.385, because flow-

only samples dominate that population and carry no application-layer text to match, with T1110 recording a hit@3 of 0.00 across 236 queries. The two values describe different populations and are not interchangeable.

Evaluating only final verdicts would assign zero hallucinations to any shield, including an entirely inert one, so the measurement operates on execution traces recording the action before and after the shield. Of 1,566 batches, 1,421 asserted a technique, and none of those 1,421 was unanchored, a result that holds even when relaxed to the parent technique. The shield is demonstrably active in achieving this: it intervened on 9.26% of batches and, in 76 of them, downgraded BLOCK_IP to AWAIT_HITL, preventing 76 automatic permanent address blocks whose justification could not be anchored. One boundary is stated proactively: the score applies the strict criterion that the identifier appear among retrieved document identifiers, whereas the deployed shield matches by regular expression across the full context, making zero an upper bound on leakage rather than an exact count.

Tier 1 escalated 1,066 alerts, of which only 80, or 7.5%, were genuine threats. Evaluated strictly as a binary classifier over this queue, the model exhibits minimal discriminative utility: it confirmed all 1,066 events without dismissing any of the 986 benign alerts, yielding MCC 0.0, Balanced Accuracy 0.5, and a 99.55% False Positive Rate within the subset it actively confirmed. The operational value resides in the channel the binary framing disregards. The agent deferred 168 alerts, 15.76% of the input, and that queue contains 76 of the 80 genuine threats, or 95.0%, at a true-alert density of 45.24% against 7.5% at the input: a $6.03\times$ enrichment, such that an analyst reviewing only the deferred queue examines 84.24% less material while still encountering 95.0% of the real threats. The model therefore operates as a triage router rather than an autonomous decision authority; and since 910 of the 1,066 escalations were benign CSIC traffic, the figure also reflects the Tier 1 escalation decision rather than the behaviour of the agent alone.

A cross-family judge, Meta-Llama-3-8B-Instruct scoring Foundation-Sec-8B-Instruct, graded all 1,052 escalated events on the four RAGAS axes [40], following the recommended protocol against self-enhancement bias [41]: context precision 2.54, answer rele-

Tier 2 alert triage: the value is in the deferral channel, not the verdict

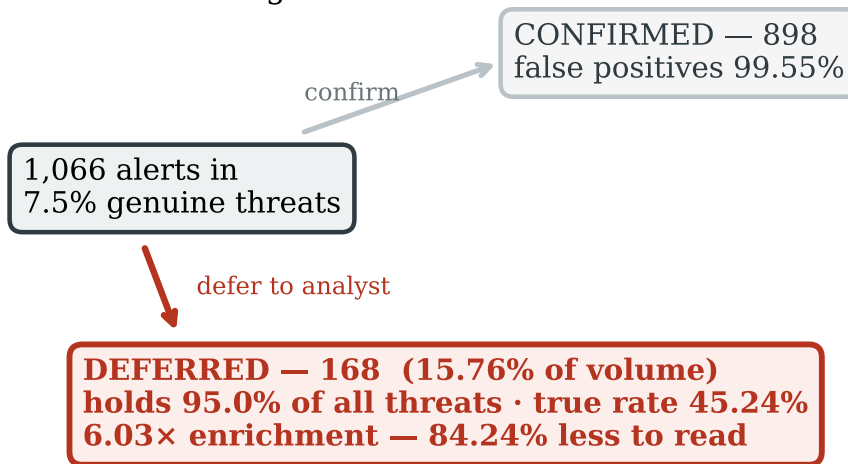


Figure 9. Alert triage at Tier 2, contrasting the confirmation channel with the deferral channel. Generated from `tier2_decision_results.json`. (Source: Author)

vancy 4.52, faithfulness 3.96, and context recall 4.11, for a mean of 3.78 out of 5.0. Context precision is the weakest axis and corroborates the retrieval diagnosis, since the retrieved set frequently contains entries irrelevant to the event. More consequentially, evidence grounding reaches only 11.2% (95% CI 9.5–13.3), so approximately nine justifications in ten quote no log value verifiable against the record. This is the widest gap between what the architecture guarantees and what it volunteers: anchoring proves that an asserted identifier is supported, but nothing compels the surrounding prose to cite the field values behind it, so transparent forensic reporting is only partially achieved.

The component ablation was scored on the final action rather than on a binary verdict, because at an 86.9% attack base rate every configuration scores close to that figure and the comparison collapses; the emitted action was instead compared against the expected one across BLOCK_IP, ALERT, LOG and AWAIT_HITL. The change of measure is disclosed because post-hoc metric selection warrants methodological scrutiny, its justification being the statistical degeneracy of the binary measure on this distribution rather than its unfavourable outcome. On 1,700 samples the complete two-tier system attains 35.35% action accuracy (95% CI 33.12–37.66) against 28.29% (95% CI 26.20–30.48) for Tier 1 alone, with non-overlapping intervals. The trade-off nevertheless inverts: Tier 1 alone is the more reliable adjudicator whenever it commits, at 69.01% autonomous precision against 57.72%, yet it commits on only 41.00% of events and leaves 59.00% unresolved, escalating them to a tier

that configuration A does not possess. The complete system leaves nothing unresolved, converting that backlog into 44.76% explicit deferrals alongside 55.24% autonomous decisions. The two configurations therefore differ less in the quality of their reasoning than in the action they execute under uncertainty, the same property the deferral channel measures from the opposite direction.

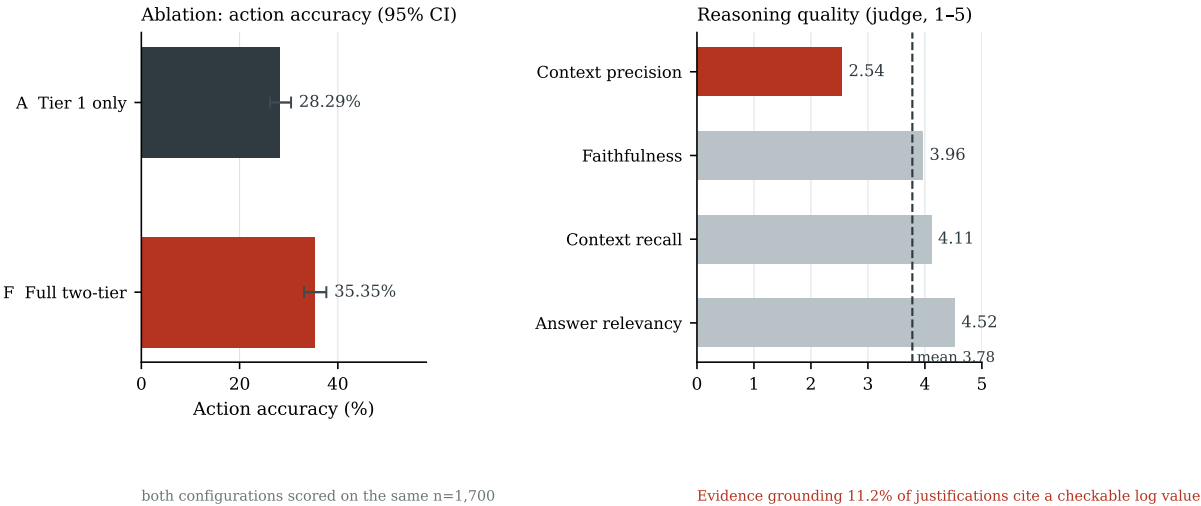


Figure 10. Left: action accuracy of Tier 1 alone against the complete two-tier system, with 95% confidence intervals. Right: the four judge axes, with context precision the weakest. Generated from `ablation_action_scores.json` and `reasoning_eval_results.json`. (Source: Author)

Chapter 5

Conclusion

A Security Operations Centre rarely fails because an intrusion went undetected. It fails at the step after detection, where a finite number of specialists must decide which of an effectively unbounded stream of alerts deserves their attention. Language models are the first technology capable of reading an event in its context and explaining it in words a person can act on, yet placing one in an operational stream imposes costs in time, trust and data sovereignty that no headline accuracy figure addresses. This thesis asked whether that capability can be obtained without paying its full price, and answered by building SENTINEL—a two-tier architecture that settles cheap decisions before expensive ones, deployed entirely on a local workstation—and then measuring it rather than describing it. Three results carry the claim.

The economics of tiering hold as a systematic principle. Tier 1 (combining application signature Rule-based rules and Welford statistical filtering) and the Machine Learning Gate resolve 90.6% of a benchmark stream carrying more than three times the realistic attack proportion, and 97.5% of a stream at a near-realistic base rate, in both cases without a single language model invocation. Because the typical event never reaches the model, median end-to-end latency falls to 0.88 ms from 17,174.7 ms under a single-tier baseline. Methodologically, the offload rate decomposes linearly into a benign-case and an attack-case rate, making it a direct function of the stream’s attack base rate rather than a static system constant.

The security guarantee is structural, and therefore does not decay. Delimited Data Encapsulation treats content inside a per-batch nonce boundary as passive text regardless of what it says, and resisted all 678 adversarial samples in the categories the inexpensive static pre-

filter cannot absorb—precisely the categories that filter is blind to. This is a different kind of assurance from a classifier that blocks $x\%$ and leaks the remainder against every novel phrasing. HMAC-SHA256 chaining detects 100% of the three core tampering modes with no false alarm over an intact control chain, sealing the forensic record with integrity and tamper-evidence.

Hallucination control became a verifiable constraint rather than a hope. Every ATT&CK identifier the system emits must appear in that batch's retrieved documents; across 1,421 adjudications asserting a technique, zero were unanchored. That zero is not the trivial score of a shield that never acts: it intervened on 9.26% of batches, and 76 of those interventions withheld a permanent IP block the system could not justify from evidence. The reasoning tier also proved most valuable in a role other than the obvious one—used as a triage router rather than an autonomous confirmer, its deferral queue concentrates 95.0% of genuine threats into 15.76% of the volume, a $6.03\times$ enrichment that removes 84.24% of the material an analyst must read.

The architecture's limits are reported alongside the measurements that expose them, in Chapter 4, and three set the immediate agenda: attribution is currently better without the reasoning tier than with it, the agent cannot yet dismiss a false alarm on its own, and the explanation layer is the weakest link, since retrieval admits irrelevant material and only 11.2% of justifications cite a log value that can be checked back against the record. The corresponding directions are a retrieval layer repaired at the knowledge-base level, calibration for dismissal as well as confirmation, and a justification format that compels citation rather than merely permitting it. Two smaller extensions follow from properties already stated: anchoring hash values periodically outside the system would close the tail-truncation blind spot inherent to log chaining, and asymmetric signatures would upgrade tamper-evidence to genuine non-repudiation.

Concluding a rigorous research endeavor, while opening inspiring horizons in securing the digital realm through the power of Agentic Artificial Intelligence.

References

- [1] G. González-Granadillo, S. González-Zarzosa, and R. Diaz, “Security Information and Event Management (SIEM): Analysis, Trends, and Usage in Critical Infrastructures,” *Sensors*, vol. 21, no. 14, art. 4759, 2021.
- [2] S. Tariq, M. Baruwal Chhetri, S. Nepal, and C. Paris, “Alert Fatigue in Security Operations Centres: Research Challenges and Opportunities,” *ACM Computing Surveys*, vol. 57, no. 9, art. 224, 2025.
- [3] B. A. AlAhmadi, L. Axon, and I. Martinovic, “99% False Positives: A Qualitative Study of SOC Analysts’ Perspectives on Security Alarms,” in *Proceedings of the 31st USENIX Security Symposium*, 2022, pp. 2783–2800.
- [4] Ismail, R. Kurnia, Z. A. Brata, G. A. Nelistiani, S. Heo, H. Kim, and H. Kim, “Toward Robust Security Orchestration and Automated Response in Security Operations Centers with a Hyper-Automation Approach Using Agentic Artificial Intelligence,” *Information*, vol. 16, no. 5, art. 365, 2025.
- [5] A. Vaswani *et al.*, “Attention is All You Need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, pp. 5998–6008, 2017.
- [6] K. Greshake, S. Abdelnabi, S. Mishra, A. Endres, T. Holz, and M. Fritz, “More than you’ve asked for: A Comprehensive Analysis of Novel Prompt Injection Threats to Application-Integrated Large Language Models,” *arXiv preprint arXiv:2302.12173*, 2023.
- [7] S. Rose, O. Borchert, S. Mitchell, and S. Connelly, “Zero Trust Architecture,” NIST Special Publication 800-207, National Institute of Standards and Technology, 2020.
- [8] OWASP GenAI Security Project, “OWASP Top 10 for Large Language Model Applications,” Version 2025, The OWASP Foundation, 2025. [Online]. Available: <https://genai.owasp.org/>
- [9] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, “Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization,” in *Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP)*, 2018, pp. 108–116.
- [10] C. Torrano-Giménez, A. Pérez-Villegas, and G. Álvarez Marañón, “HTTP DATASET CSIC 2010,” Information Security Institute, Spanish National Research Council (CSIC), 2010. [Online]. Available: <https://www.isi.csic.es/dataset/HTTP>
- [11] A. Zou, Z. Wang, J. Z. Kolter, and M. Fredrikson, “Universal and Transferable Adversarial Attacks on Aligned Language Models,” *arXiv preprint arXiv:2307.15043*, 2023.
- [12] Deepset, “Prompt Injections Dataset,” *Hugging Face repository*, 2023. [Online]. Available: <https://huggingface.co/datasets/deepset/prompt-injections>
- [13] jackhhao, “Jailbreak Classification Dataset,” *Hugging Face repository*, 2023. [Online]. Available: <https://huggingface.co/datasets/jackhhao/jailbreak-classification>
- [14] R. Zuech, T. M. Khoshgoftaar, and R. Wald, “Intrusion Detection and Big Heterogeneous Data: A Survey,” *Journal of Big Data*, vol. 2, art. 3, 2015.
- [15] B. P. Welford, “Note on a Method for Calculating Corrected Sums of Squares and Products,” *Technometrics*, vol. 4, no. 3, pp. 419–420, 1962.

- [16] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “Light-GBM: A Highly Efficient Gradient Boosting Decision Tree,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [17] A. Gholami *et al.*, “A Survey of Quantization Methods for Efficient Neural Network Inference,” *arXiv preprint arXiv:2103.13630*, 2021.
- [18] S. Weerawardhena, P. Kastianik, B. Nelson, B. Saglam *et al.* (Cisco Foundation AI), “Llama-3.1-FoundationAI-SecurityLLM-8B-Instruct Technical Report,” *arXiv preprint arXiv:2508.01059*, 2025.
- [19] P. Kastianik, B. Saglam *et al.* (Cisco Foundation AI), “Llama-3.1-FoundationAI-SecurityLLM-Base-8B Technical Report,” *arXiv preprint arXiv:2504.21039*, 2025.
- [20] LangChain AI, “LangGraph: Multi-Actor, Stateful LLM Applications,” *GitHub repository*, 2024. [Online]. Available: <https://github.com/langchain-ai/langgraph>
- [21] P. Lewis *et al.*, “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.
- [22] J. Johnson, M. Douze, and H. Jégou, “Billion-Scale Similarity Search with GPUs,” *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
- [23] S. Robertson and H. Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond,” *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.
- [24] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, “Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods,” in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2009, pp. 758–759.
- [25] B. E. Strom, A. Applebaum, D. P. Miller, K. C. Nickels, A. G. Pennington, and C. B. Thomas, “MITRE ATT&CK: Design and Philosophy,” The MITRE Corporation, Technical Report MP180360, 2018.
- [26] P. Cichonski, T. Millar, T. Grance, and K. Scarfone, “Computer Security Incident Handling Guide,” NIST Special Publication 800-61 Revision 2, 2012.
- [27] R. Pandey and A. Bhujang, “Poisoning the Watchtower: Prompt Injection Attacks Against LLM-Augmented Security Operations Through Adversarial Log Content,” *arXiv preprint arXiv:2605.24421*, 2026.
- [28] H. Krawczyk, M. Bellare, and R. Canetti, “HMAC: Keyed-Hashing for Message Authentication,” Internet Engineering Task Force, RFC 2104, 1997.
- [29] J. Zhang *et al.*, “When LLMs Meet Cybersecurity: A Systematic Literature Review,” *Cybersecurity*, vol. 8, 2025.
- [30] P. He, J. Zhu, S. He, J. Li, and M. R. Lyu, “Drain: An Online Log Parsing Approach with Fixed Depth Tree,” in *Proceedings of the IEEE International Conference on Web Services (ICWS)*, 2017, pp. 33–40.
- [31] F. Blefari, C. Cosentino, F. A. Pironti, A. Furfaro, and F. Marozzo, “CyberRAG: An Agentic RAG cyber attack classification and reporting tool,” *Future Generation Computer Systems*, vol. 176, art. 108186, 2026.
- [32] Z. Liu and A. Anwar, “AutoBnB-RAG: Enhancing Multi-Agent Incident Response with Retrieval-Augmented Generation,” *arXiv preprint arXiv:2508.13118*, 2025.
- [33] A. Abdennebi, N. Kara, L. Lahlou, and H. Ould-Slimane, “LanG – A Governance-Aware Agentic AI Platform for Unified Security Operations,” *arXiv preprint arXiv:2604.05440*, 2026.
- [34] R. Sahay *et al.*, “Policy-Guided Threat Hunting: An LLM Enabled Framework with

- Splunk SOC Triage,” *arXiv preprint arXiv:2603.23966*, 2026.
- [35] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, “A Design Science Research Methodology for Information Systems Research,” *Journal of Management Information Systems*, vol. 24, no. 3, pp. 45–77, 2007.
 - [36] OWASP Foundation, “OWASP Top 10:2021 - The Ten Most Critical Web Application Security Risks,” The OWASP Foundation, 2021. [Online]. Available: <https://owasp.org/Top10/>
 - [37] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” in *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
 - [38] S. Myneni, A. Chowdhary, A. Sabur, S. Sengupta, G. Agrawal, D. Huang, and M. Kang, “DAPT 2020 – Constructing a Benchmark Dataset for Advanced Persistent Threats,” in *Deployable Machine Learning for Security Defense (MLHat 2020)*, Communications in Computer and Information Science, vol. 1271, Springer, 2020, pp. 138–163.
 - [39] X. Shen, Z. Chen, M. Backes, Y. Shen, and Y. Zhang, “‘Do Anything Now’: Characterizing and Evaluating In-The-Wild Jailbreak Prompts on Large Language Models,” in *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2024, pp. 1671–1685.
 - [40] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, “RAGAS: Automated Evaluation of Retrieval Augmented Generation,” *arXiv preprint arXiv:2309.15217*, 2023.
 - [41] L. Zheng *et al.*, “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.